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NIYONKURU JACQUES

PROFESSIONAL SUMMARY

Full-stack JavaScript/TypeScript software engineer with experience building web applications using Node.js, Express, React, GraphQL, MongoDB, and PostgreSQL. Skilled in writing clean, maintainable code, debugging complex issues, and improving existing implementations in collaborative Git-based environments.

Prior to transitioning into software development, I worked for **seven years as a Nurse-Midwife in public hospitals in Rwanda**, delivering maternal and community healthcare services. This frontline healthcare experience inspired my transition into technology and fuels my commitment to developing digital solutions that improve healthcare systems and enhance people's lives.

I bring a unique combination of **clinical insight and technical expertise**, with a growing interest in **AI systems and LLM-based applications** focused on improving real-world problem solving and system efficiency.

TECHNICAL SKILLS

- **Languages:** JavaScript (Advanced), TypeScript (Intermediate), Python (Basic)
- **Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Material UI, Redux
- **Backend:** Node.js, Express.js, FastAPI, GraphQL, REST APIs
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Tools & Workflow:** Git, GitHub, VS Code, Docker (basic), CI/CD basics
- **Core Skills:** Code Review, Debugging, Clean Code, API Design, Problem Solving, Edge Case Handling, Software Testing
- **AI / Systems:** LangGraph, MCP architecture, LLM-based agents

PROFESSIONAL EXPERIENCE

Full-stack software developer (self driven projects)

Developed multiple full-stack applications using modern JavaScript/TypeScript ecosystem tools.

2023 - PRESENT

- Built scalable web applications using React, Node.js, and Express
- Designed and implemented REST and GraphQL APIs
- Developed and managed database schemas using MongoDB and PostgreSQL
- Debugged and fixed issues in frontend and backend systems
- Reviewed and improved existing code for performance and readability
- Applied Git workflows for version control and collaboration
- Ensured clean, reusable, and maintainable code structure

PROJECTS

Expenses tracker AI agent

CURRENT

An AI-assistant that lets you log expenses using natural language or voice.

Chat interactions are triggered from an MCP client built with copilot-kit, communicating with an MCP server that orchestrates tool calls via llm-powered LangGraph.

Tech Stack: FastAPI, MySQL, CopilotKit, LangGraph, Gemini

Key Features:

- Natural language and voice-based expense tracking
- AI agent for reasoning and tool orchestration
- Modular MCP client-server architecture
- LLM-based decision-making workflow

Highlights:

- Designed and implemented AI agent workflow using LangGraph
- Built tool-based architecture for structured LLM interactions
- Focus on correctness, reasoning flow, and system reliability

Community Volunteering Management System

FINAL YEAR PROJECT

Full-Stack Web Application

A platform designed to connect community organizations with skilled volunteers for social impact initiatives.

Tech Stack: React, Node.js, Express, MongoDB

Key Features:

- Volunteer registration and skill matching system
- Organization event and opportunity management
- REST API backend with structured architecture
- Responsive React frontend

Highlights:

- Implemented full-stack architecture from scratch
- Focused on scalability, maintainability, and clean code design
- Strong emphasis on backend logic and data modeling

EDUCATION

University of Kigali, Rwanda – Bachelor of computer sciences

March 2021 – August 2025, RWANDA

- Fourth best performer in the department of **computer sciences**.

University of Rwanda, Rwanda – Advanced diploma in nursing sciences and midwifery

October 2013 – July 2016, RWANDA

PROFESSIONAL TRAINING

Software engineering trainee – Andela(2023)

- Full-stack JavaScript development
- Clean code principles and best practices
- Git workflows and collaborative development
- Debugging and problem-solving techniques
- Building production-ready applications

KEYS STRENGTHS

- Strong JavaScript and TypeScript fundamentals
- Experienced in debugging and improving code quality
- Skilled in code review, edge case detection, and correctness validation
- Ability to analyze and refine technical solutions
- Experience working with AI agent systems and LLM-based workflows
- Strong attention to detail and structured thinking
- Comfortable working independently in remote environments
- Interest in evaluating AI-generated code and improving output quality

LANGUAGES

- English (proficient)
- French (advanced)
- Kinyarwanda(native)